

DSP VERSION 1.0

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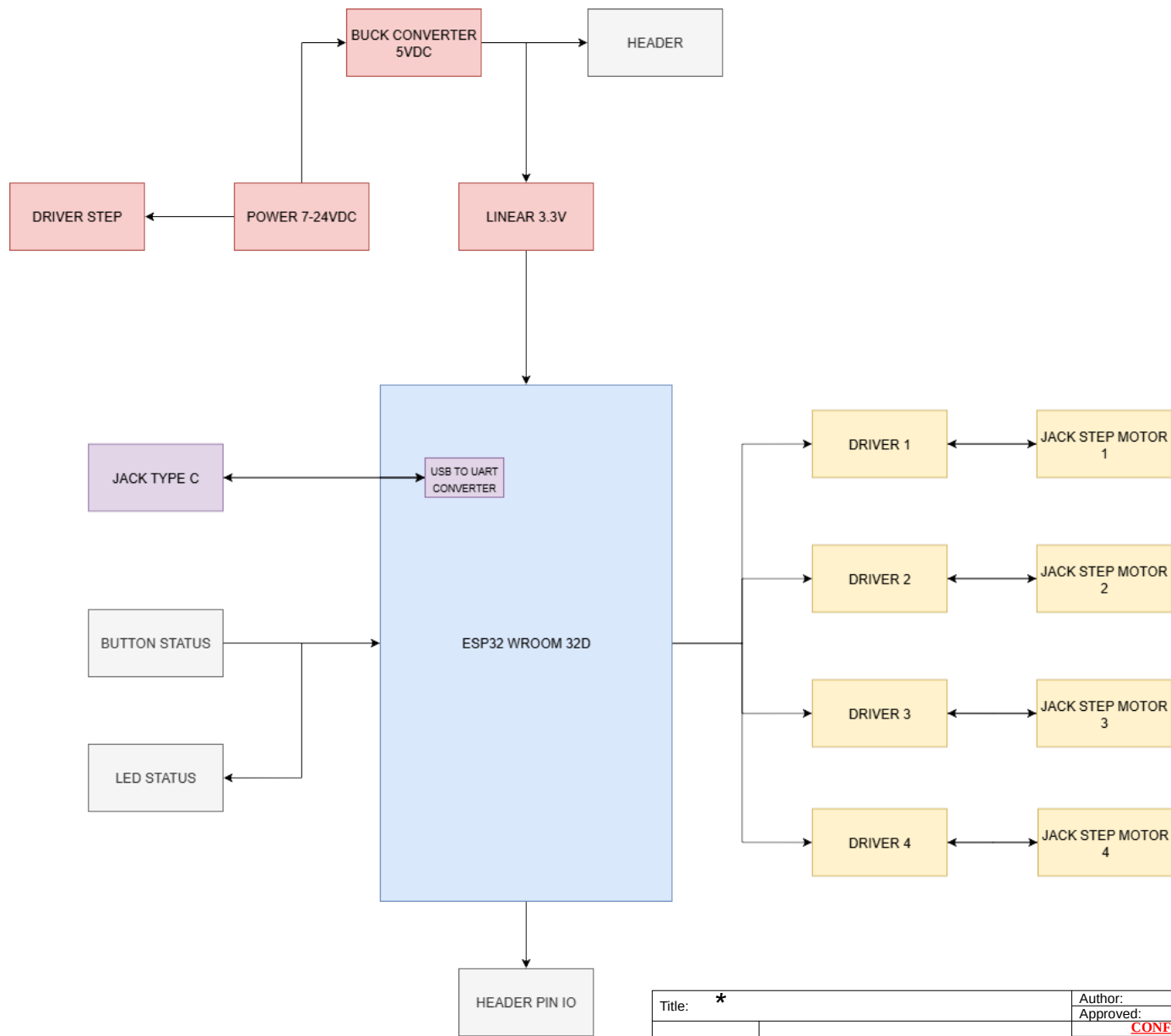
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DESIGN CONSIDERATIONS

DESIGN NOTE:
Functions of this block

DESIGN NOTE:
Calculation equations

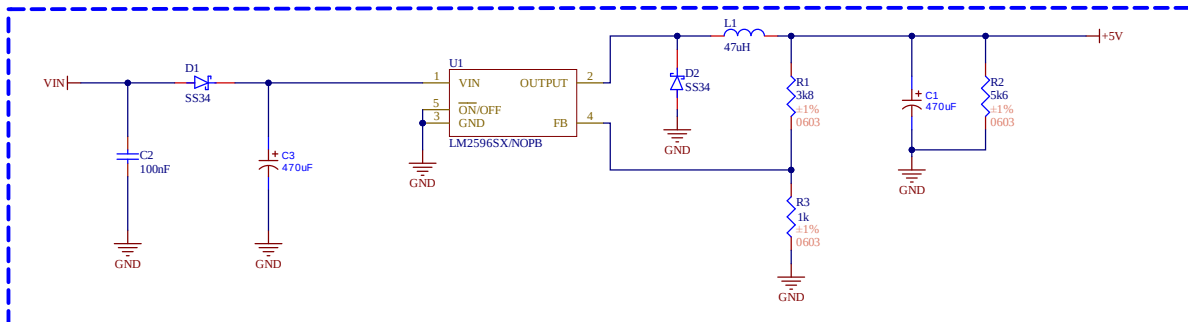
DESIGN NOTE:
Calculations when choosing
this component



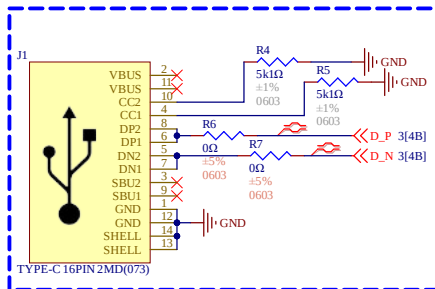
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Size: A4			Approved: *
Project: DSP_VERSION_1.0			CONFIDENTIAL
Date: 4/7/2026			Modified Date: 4/6/2026
11:19:17 PM			Revision: *
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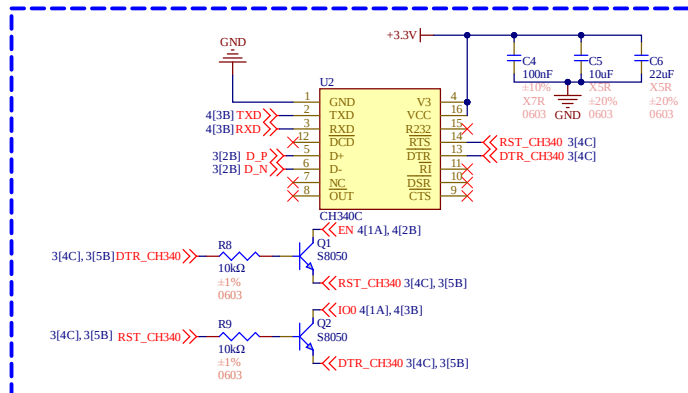
BUCK CONVERTER



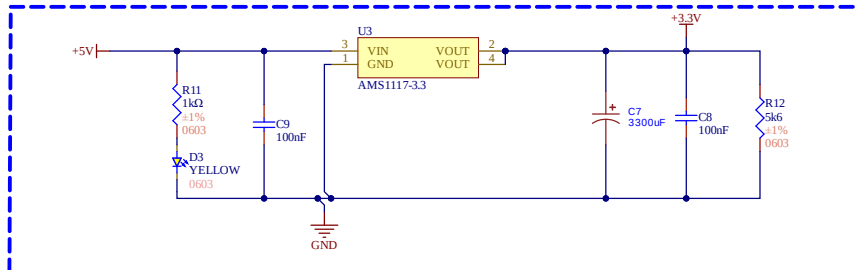
USB TYPE C



USB TO UART CONVERTER

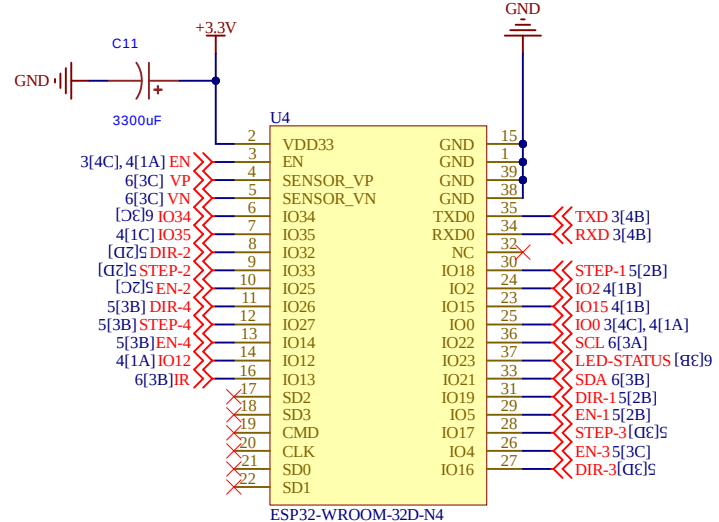
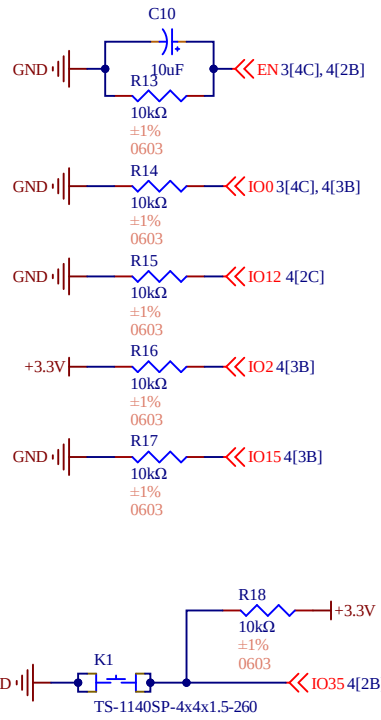


LINEAR MCU



Title: *	Author: *
Size: A3	Project: DSP_VERSION_10
Date: 4/7/2026	Modified By: *
11:19:17 PM	Modified Date: 4/7/2026
Sheet 3 of 6	Revision: *
	SW Version: 24.1.2.44

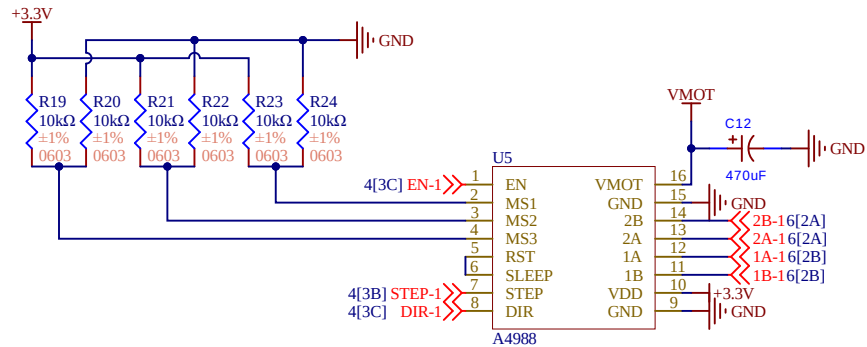




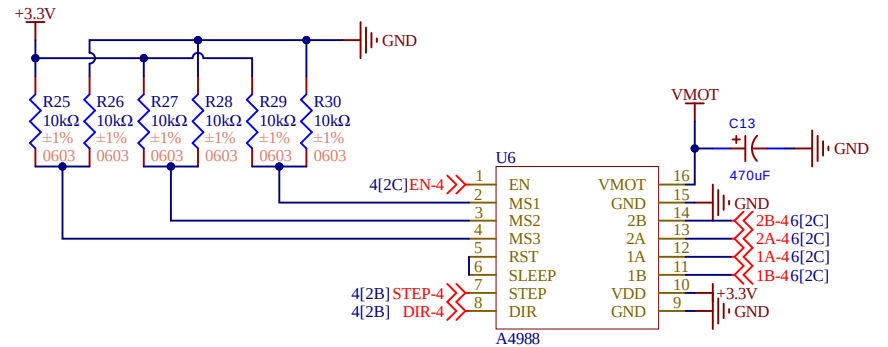
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			Approved: *
Size: A4	Project: DSP_VERSION_1.0		CONFIDENTIAL
Date: 4/7/2026	11:19:18 PM	Sheet 4 of 6	Modified Date: 4/7/2026
			Revision: *
			SW Version: 24.1.2.44



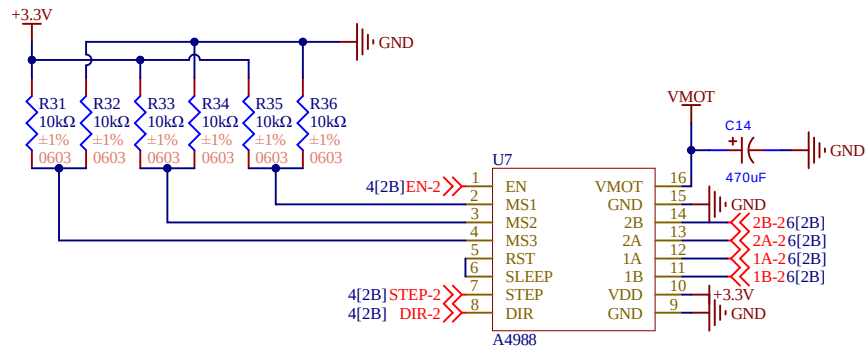
DRIVER STEP 1



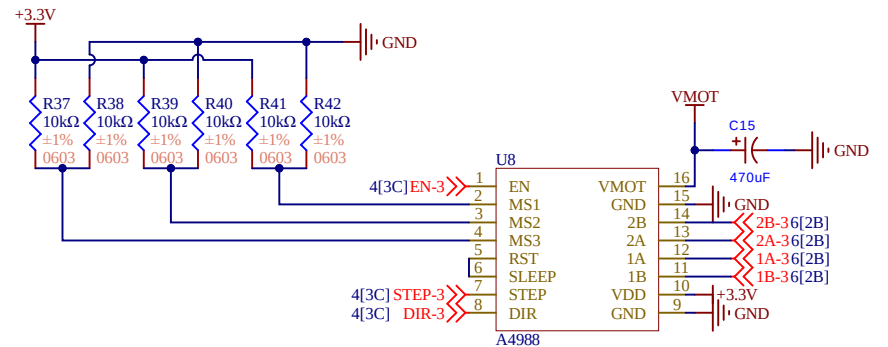
DRIVER STEP 4



DRIVER STEP 2

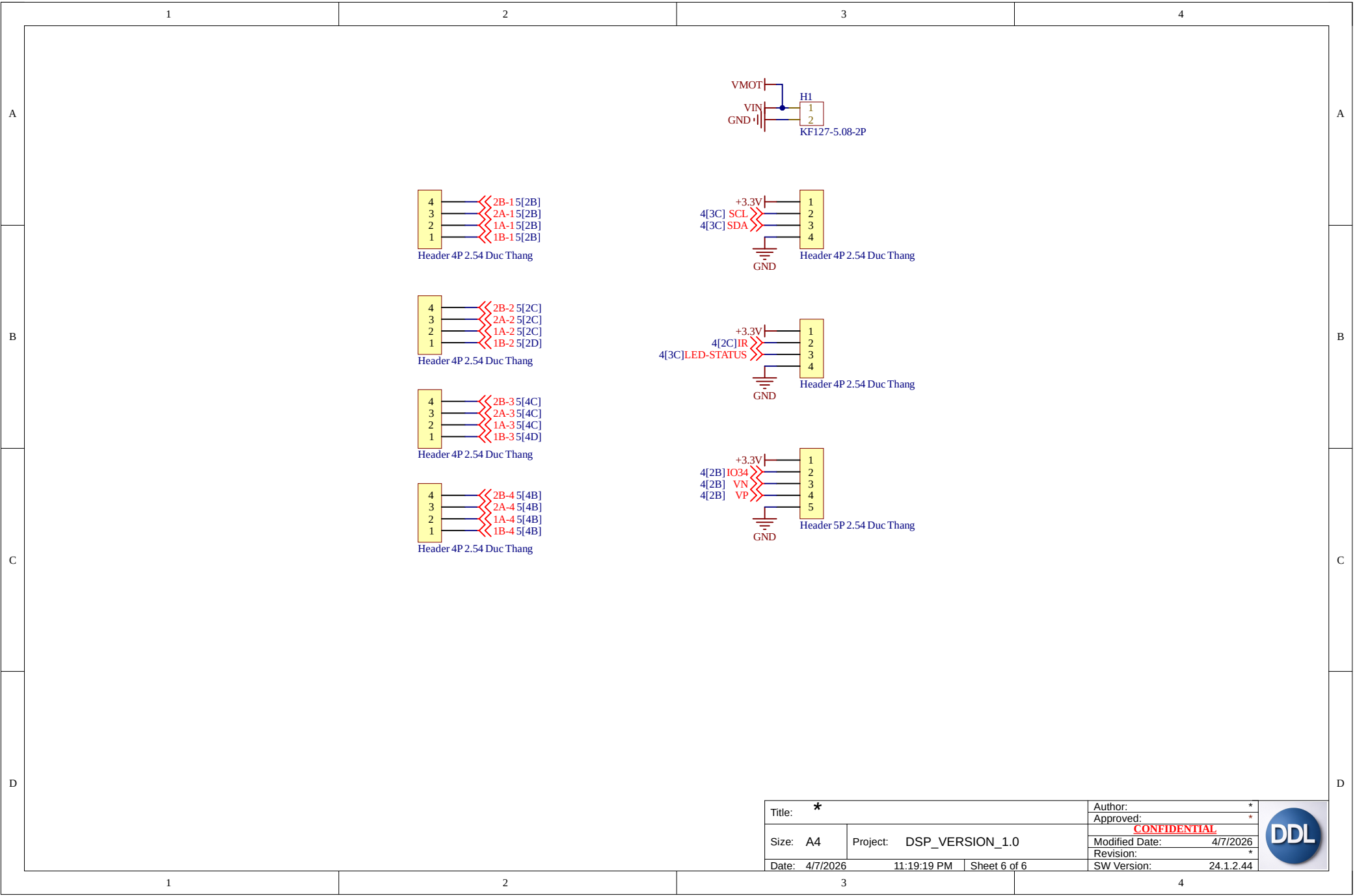


DRIVER STEP 3



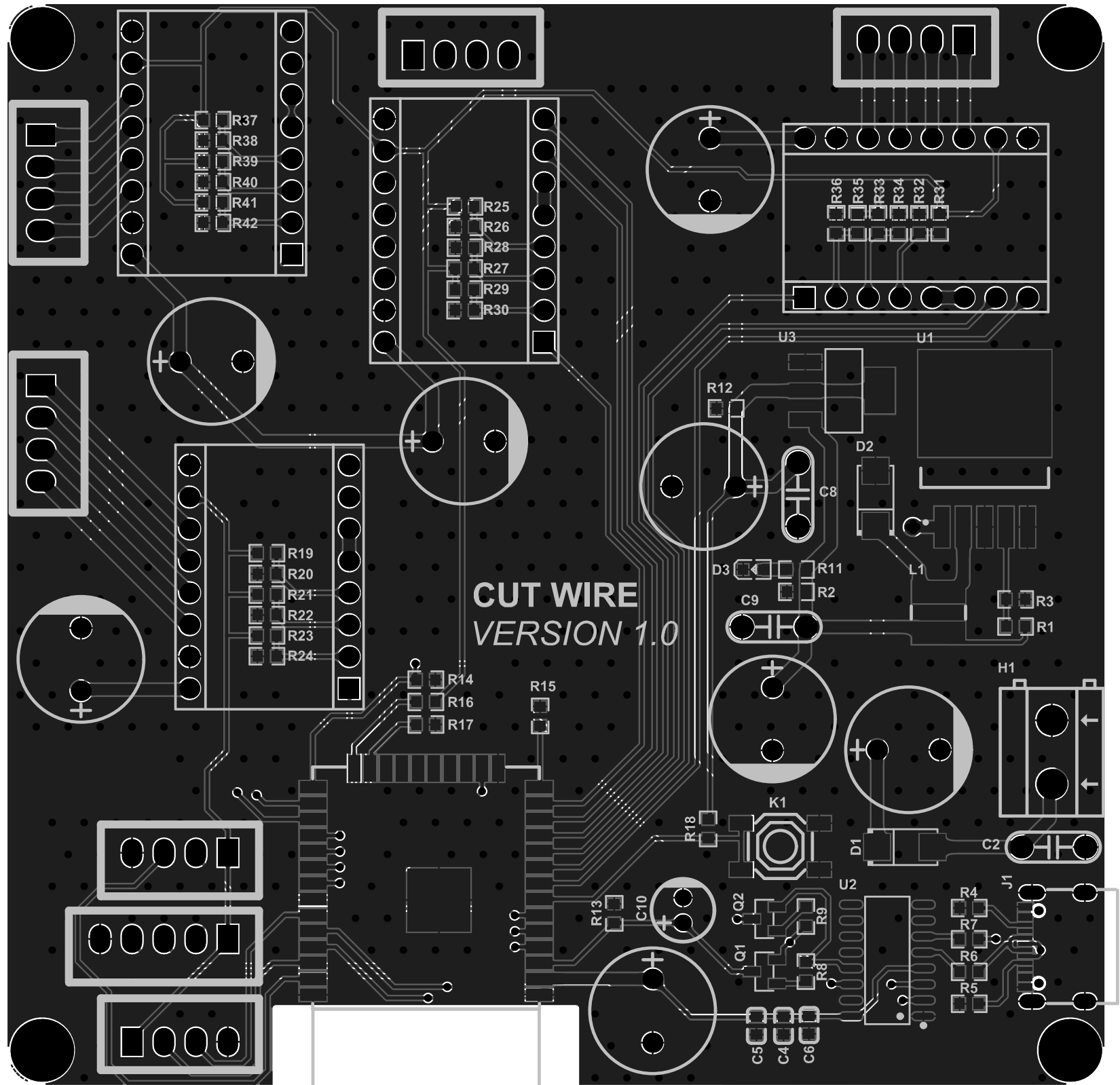
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Project: DSP_VERSION_1.0			CONFIDENTIAL	
Date: 4/7/2026			Modified Date: 4/7/2026	
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Title: *			Author: *
			Approved: *
Size: A4	Project: DSP_VERSION_1.0		Modified Date: 4/7/2026
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Comment	Description	Designator	Footprint	LibRef	Quantity
470uF	470uF ±20% 35V Through Hole D10xL13mm	C1, C3, C12, C13, C14, C15	470uF_35V, D10xL13mm	470uF_35V, D10xL13mm	6
100nF	C4m	C2, C8, C9	tu tantalum dp5.08	Tu tantalum 100nF 50V 25% DIPS.08	3
100nF	50V 100nF X7R ±10% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C4	C0603	C0603_100nF_±10%_ X7R	1
10uF	25V 10uF X5R ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C5	C0603	C0603_10uF_±10%_X 7R	1
22uF	6.3V 22uF X5R ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C6	C0603	C0603_22uF_±20%_X 5R	1
3300uF	3300uF ±20% 6.3V Through Hole D8xL16mm 2000hrs	C7, C11	Through Hole D8xL16mm	Through Hole D8xL16mm	2
10uF	10uF 50V ±20% Through Hole D5xL11mm Aluminum Electrolytic Capacitors - Leaded ROHS	C10	C_TH_5x11mm	C_10uF_5x11TH	1
SS34	40V Independent 3A 550mV@3A 3MAA(DO- 214AC) Schottky Diodes ROHS	DL, D2	SS34	SS34	2
YELLOW	25mcd-40mcd Water Clear -40°C~+85°C 634nm Red 120° 60mW 0402 LED Indication - Discrete ROHS	D3	LED0603	D0603_YELLOW	1
KF127-5.08-2P	1.5 xDP 14-26 Green 40°C~+105°C 15A 250V Through Hole 5.08mm 1 2 Through Hole,P=5.08mm Wire to Board ROHS	H1	KF127-5.08-2P	KF127-5.08-2P	1
Header 4P 2.54 Duc Thang	C4m	H3, H4, H5, H6, H7, H9	XH2.54M-4	Header 4P 2.54 Duc Thang	6
Header 5P 2.54 Duc Thang	C4m	H8	XH2.54M-5	Header 5P 2.54 Duc Thang	1
TYPE-C 16PIN 2MD(073)	3A 1 Surface Mount, Right Angle 16P Female -25°C~+85°C Type-C SMD USB Connectors ROHS	J1	TYPE_C_FAMALE	TYPE_C_FAMALE_16P IN	1
TS-1140SP-4x4x1.5- 260	SMD Tactile Switches ROHS	K1	4x4x1.5	TS-1140SP-4x4x1.5- 260	1
47uH	1.15A 10uH Unshielded Inductor ±20% SMD 4x4.5mm Power Inductors ROHS	L1	CD43	CD4-10uH	1
S8050	25V 300mW 500mA SOT-23 Bipolar (BJT) ROHS	Q1, Q2	S8050	S8050	2
3k8	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R1	R0603	R0603_4k70_±1%	1
5k6	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R2, R12	R0603	R0603_4k70_±1%	2
1k	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R3	R0603	R0603_4k70_±1%	1
5k10	100mW Thick Film Resistor 75V ±100ppm/°C ±1% 5.1k0 0603 Chip Resistor - Surface Mount ROHS	R4, R5	R0603	R0603_5k10_±1%	2
00	100mW Thick Film Resistor 75V ±5% 00 0603 Chip Resistor - Surface Mount ROHS	R6, R7	R0603	R0603_00_±5%	2
10k0	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R8, R9, R13, R14, R15, R26, R27, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42	R0603	R0603_10k0_±1%	32
1k0	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R11	R0603	R0603_1k0_±1%	1
LM2596SX/NOPB	150kHz Buck Fixed 1 TO-263 S Voltage Regulators - DC/DC Switching Regulators ROHS	U1	LM2596SX-5.0/NOPB	LM2596SX/NOPB	1
CH340C	2Mbps Transceiver USB 2.0 SOP-16 USB Converters ROHS	U2	CH340C	CH340C	1
AMS1117-3.3	72dB@1(20Hz) 1A Fixed 3.3V Positive 18V SOT-223 Voltage Regulators - Linear, Low Drop Out (LDO) Regulators ROHS	U3	AMS1117-3.3	AMS1117-3.3	1
ESP32-WROOM-32D- N4	SMD 18x25.5mm WiFi Modules ROHS	U4	ESP32-WROOM-32D- N4	ESPRESSIF ESP32- WROOM-32D-N4	1
A4988		U5, U6, U7, U8	A4988-driver_step	Module_A4988	4